

Revealing the Swan: Transforming Mutualism from Peripheral to a Core Pillar of Ecological Theory

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Abstract

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Introduction: The Ugly Duckling of Ecological Theory

For over a century, mutualism has been the “ugly duckling of community ecology” (Mittelbach and McGill, 2019). While competition and predation matured into elegant, mathematically rigorous, and empirically testable bodies of theory that now occupy core chapters in ecology textbooks, mutualism has historically remained at the periphery. Several explanations have been proposed for this persistent neglect (see Boucher et al., 1982; Bronstein, 1994; Raerinne, 2020; Bronstein, 2015; Hale and Valdovinos, 2021), including two I aim to refute in this forum piece. First, the Lotka–Volterra model for mutualism predicts unbounded growth for strong mutualisms, which discouraged theoretical work on mutualism at a time when May (1976, 1982) argued that such unbounded growth explained why mutualisms were rare in nature. Second, the diversity of mechanisms by which species benefit one another was assumed to defy a unified theory (Risch and Boucher, 1976; Boucher et al., 1982; Bronstein, 1994), thereby discouraging attempts at theoretical unification. Today, although empirical research on mutualism has blossomed (Bronstein, 1994, 2015, 2026; Raerinne, 2020) and mutualism theory has accumulated (Hale and Valdovinos, 2021), this theory remains insufficiently integrated into our general understanding of ecology.

Table 1: Coverage of mutualism, competition, and predation in ecology textbooks. Values are page counts for each interaction type. Asterisks preceding textbook names indicate books analyzed by [Keddy \(1990\)](#). Parenthetical values in the mutualism column indicate pages devoted to symbiosis where authors appeared to treat symbiosis as equivalent to mutualism, except for Vandermeer and Goldberg, where it indicates page counts in the second edition. Counts for competition and predation exclude chapters or sections on food webs and indirect effects, although those topics are about competitive and consumer-resource interactions. The Ricklefs and Miller predation value includes all consumer–resource interactions. The [Mittelbach and McGill \(2019\)](#) value combines mutualism and facilitation. [†] Includes all consumer–resource interactions.

Textbook	Mutualism	Competition	Predation
*Colinvaux (1986)	1	33	70
*Collier et al. (1973)	0 (1)	45	30
*Hutchinson (1978)	0 (9)	59	6
*Krebs (1978)	3	50	32
*Lederer (1984)	5	21	4
*McNaughton and Wolf (1979)	20	77	71
*Odum (1983)	15	17	15
*Pianka (1983)	3	74	41
*Ricklefs (1979)	3	38	30
*Ricklefs (1983)	2	11	14
*Smith (1986)	2 (1)	19	24
*Whittaker (1975)	5 (9)	18	22
Begon, Harper, and Townsend (1990)	37	39	82
Gotelli (1995)	0	26	30
Morin (1999)	21	76	51
Case (1999)	0.5	23	68
Ricklefs and Miller (2000)	9	45	78 [†]
Mittelbach and McGill (2019)	16	38	53
Vandermeer and Goldberg (2003, 2013)	5 (14)	29	38

This limited integration is reflected in the treatment of mutualism in ecology textbooks and theoretical syntheses Table 1. Here, I use mutualism to mean reciprocal net benefit between species, distinct from symbiosis, which refers to intimate association, and

facilitation, which need not be reciprocal (Bronstein, 2015). Since the 1970s, researchers have repeatedly reported the diminished treatment of mutualism with respect to competition and predation in ecology textbooks (Risch and Boucher, 1976; Boucher, 1985a; Keddy, 1990). Keddy (1990), for example, found that 12 introductory ecology textbooks devoted roughly 30–38 pages to competition and predation, but fewer than 7 pages to mutualism (Table 1). The few ecology textbooks that provide substantial discussions on mutualism, for example Begon et al. (1990) and Mittelbach and McGill (2019), mostly cover empirical examples, study systems, or specialized mechanisms of reciprocal benefit, rather than the robust theoretical framework that has accumulated over the last decades (Valdovinos, 2019; Hale and Valdovinos, 2021). This pattern persists in later theoretical syntheses: Case (1999) and Gotelli (1995) devote no dedicated sections to mutualism, and Hastings and Gross’s *Encyclopedia of Theoretical Ecology* includes only a brief entry under “facilitation.” Vandermeer and Goldberg (2003) provide the strongest theoretical treatment, expanding their mutualism coverage from 5 pages in the first edition to 14 pages in the second. Yet even this unusually strong treatment stops short of fully developing mutualism theory into a framework for generating testable predictions, guiding empirical research, and informing applied ecological questions.

This textbook pattern matters because it shapes how ecologists learn to think and conduct research. Students commonly encounter competition and predation through equations, zero-growth isoclines, equilibria, and stability analysis, and therefore learn how these interactions generate general principles such as exclusion, coexistence, oscillations, and density dependence. Mutualism, by contrast, is more often taught through examples such as pollination, seed dispersal, mycorrhizae, or defensive symbioses, with less emphasis on the population-dynamic principles that apply across these systems. This difference in pedagogy can reinforce a difference in research practice. Work on competition and predation often begins with model-based expectations and then chooses systems in which to test them, whereas work on mutualism often begins with particular systems and asks questions specific

to those systems. Important exceptions exist, especially in mutualistic network theory (Bascompte and Jordano, 2007, 2013; Valdovinos, 2019), but mutualism still lacks a commonly taught population-level framework that can guide empirical tests across systems. Thus, the marginal position of mutualism in textbooks is not only a symptom of its limited integration into theory; it may also help perpetuate it.

This forum piece argues that mutualism can be better integrated into ecology by teaching it with the same conceptual tools used for competition and predation: equations, zero-growth isoclines, equilibria, stability, and empirically testable predictions. I focus on population dynamics because they provide a simple bridge between the biological diversity of mutualisms and general ecological theory. Section 1 revisits early ecological work showing that mutualism was long recognized as ecologically important, even as its integration into general theory remained uneven. Section 2 develops a graphical synthesis of mutualism theory through Lotka–Volterra, saturating-benefit, and consumer–resource models, showing how partner dependence, bounded benefits, resource exchange, interaction costs, and density thresholds generate recurring population-dynamic predictions. Section 3 distinguishes mutualism from facilitation and symbiosis, and clarifies how environmental context can shift the balance between positive and negative interaction outcomes. I then outline empirical approaches for testing these predictions, including benefit saturation, lower partner-density thresholds, Allee effects, and upper cost-driven thresholds. The goal is not to reduce the natural-history richness of mutualisms to a single model, but to show that diverse mutualistic systems share population-dynamic features that can be visualized, modeled, and tested.

1 The Conceptual Evolution of Mutualism in Ecology

As early as the late 1960s, Eugene P. Odum argued that ecologists’ focus on antagonistic interactions provided an incomplete picture of how mature ecosystems function. Odum (1969) proposed that while early successional stages are driven by rapid growth and intense compe-

105 titution, mature ecosystems transition toward efficiency and stability through mutualism. He
106 viewed mutualism as an important evolutionary strategy for resource conservation, using the
107 Pacific coral reef as an “object lesson” of how tight mutualistic coupling between autotrophs
108 and heterotrophs allows life to persist in nutrient-poor environments (Odum, 1977). This
109 line of thinking culminated in a synthesis arguing that traditional field sampling emphasized
110 visible biomass and could therefore miss rapid, microscopic energy flows driven by mutualis-
111 tic interactions, such as root exudates feeding mycorrhizal fungi (Odum and Biever, 1984).
112 This synthesis suggested that these otherwise overlooked flows could account for more than
113 15% of net primary production and introduced them into food-web models as an “active
114 extraction” pathway. In this view, Odum treated mutualism as part of the regulatory struc-
115 ture of ecosystems, arguing that these positive feedback loops could help stabilize ecosystem
116 processes.

117 While Odum highlighted the role of mutualism in ecosystem energy flows, Douglas H.
118 Boucher worked to develop a population- and community-level theory of mutualism (Risch
119 and Boucher, 1976; Vandermeer and Boucher, 1978; Boucher et al., 1982; Boucher, 1985b).
120 During the 1970s and 1980s, ecological modeling was primarily focused on antagonistic in-
121 teractions, with mutualism often characterized as a rare or mathematically unstable phe-
122 nomenon (May, 1976). Boucher and his collaborators offered a different perspective through
123 comprehensive syntheses (Boucher et al., 1982; Boucher, 1985a) and mathematical analyses
124 (Vandermeer and Boucher, 1978; Boucher, 1985b). They argued that positive interactions
125 are both widespread in nature and mathematically tractable within existing ecological frame-
126 works. By showing that mutualism could be modeled using tools comparable to those used
127 for competition and predation, Boucher laid the groundwork for a formalized theory of posi-
128 tive interactions, even though the broader discipline was slow to integrate these models into
129 mainstream curricula.

130 Concurrently with these theoretical advances, empirical ecologists began recognizing
131 the structural importance of mutualism in maintaining biodiversity. Gilbert (1980) intro-

duced the concept of “keystone mutualists,” which are typically plants that provide disproportionate resource support to a wide array of mobile species, such as pollinators and seed dispersers. By acting as central nodes in ecological communities, these species facilitate important reproductive processes across multiple taxa. Gilbert cautioned that the loss of these highly connected mutualists could trigger cascading extinctions and declines in species richness. This work highlighted that mutualistic interactions were not only important for isolated pairs of interacting species, but also important components of community structure and resilience.

Despite the early insights established by Odum, Boucher, and Gilbert, the broader integration of mutualism into ecological theory remained limited. By the 1990s, the field had produced a growing body of empirical studies, yet it remained conceptually fragmented. [Bronstein \(1994\)](#) highlighted this disconnect, noting that empirical research was largely restricted to descriptive studies. Research efforts were heavily skewed toward specific taxonomic groups, primarily plant-pollinator and plant-frugivore interactions, and often relied on unilateral approaches that evaluated fitness consequences for only one interacting partner. Because empirical studies and mathematical modeling advanced largely in isolation from one another, research questions rarely extended beyond identifying mutualists or quantifying basic natural history. Without a conceptual synthesis to bridge these independent case studies, Bronstein argued that the field lacked the “overall picture” necessary to reveal broader ecological and evolutionary generalizations.

Over the past three decades, mutualism research has expanded substantially. As [Bronstein \(2026\)](#) outlines, the field has moved beyond isolated descriptive studies by incorporating network theory, phylogenetics, and dynamic consumer-resource models. These advances have helped clarify questions about interaction asymmetry and coevolutionary stability, demonstrating that mutualisms can be highly persistent. Yet, while these empirical and computational tools have broadened the field, translating this biological complexity into a unified theoretical framework remains an important next step.

2 Revealing the Swan: A Population-Dynamic Synthesis for Mutualism

Hale and Valdovinos's (2021) synthesis of 90 years of two-species mutualism models shows that mutualism theory is more coherent than its textbook treatment often implies. It also addresses the two explanations for mutualism's historical marginalization introduced above: the view that unbounded growth made early mutualism theory biologically suspect, and the assumption that mechanistic diversity made theoretical unification impossible. The synthesis can be developed through three increasingly mechanistic steps. Linear Lotka–Volterra mutualism first shows that unbounded growth is not the central prediction of mutualism theory, but the consequence of representing benefits as indefinitely increasing functions of partner density; its phase-plane geometry reveals how mutualisms increase coexistence densities and generate low-density thresholds. Saturating-benefit models then add the biological limits needed to replace runaway growth with finite coexistence while retaining those thresholds. Consumer-resource models finally show that mechanistic diversity does not prevent synthesis, because saturating benefits can be derived from measurable processes of resource production, reward depletion, consumption, service delivery, and demographic conversion. Across diverse derivations and empirical systems, these models repeatedly converge on stable coexistence, low-density thresholds, and, when costs or overexploitation are included, upper-density thresholds. Together, these dynamics provide a teachable framework for mutualism as population-dynamic theory and lead directly to testable predictions.

2.1 Revealing Mutualism in the Lotka–Volterra Phase Plane

A useful starting point for teaching mutualism is the Lotka–Volterra framework commonly used to introduce competition and predation. Although biologically limited because it can predict unbounded growth, the model remains valuable because it shows the core qualitative dynamics of mutualism: persistence when partners are sufficiently abundant and collapse

when partner densities fall below critical thresholds. It also shows how partner dependence and interaction strength affect zero-growth isoclines, equilibria, and density thresholds, all within the same phase-plane framework used to teach competition and predation.

For two species with densities N_1 and N_2 , the model can be written as:

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{K_1 - N_1 + \alpha_{12} N_2}{K_1} \right) \quad (1)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{K_2 - N_2 + \alpha_{21} N_1}{K_2} \right). \quad (2)$$

Here, r_i is the intrinsic growth rate of species i , K_i is the density species i can maintain in the absence of its partner, and α_{ij} measures the positive effect of species j on species i . This formulation is the same as the Lotka–Volterra competition model, except that the interspecific terms are positive rather than negative. The zero-growth isoclines make the contrast with competition especially clear. Setting each growth equation to zero gives:

$$N_1 = K_1 + \alpha_{12} N_2 \quad (3)$$

$$N_2 = K_2 + \alpha_{21} N_1. \quad (4)$$

These upward-sloping isoclines show that each species can persist above its single-species carrying capacity when its partner is present, and that this partner-supported density increases as partner abundance increases (Figure 1). They also provide a clear contrast with competition, where isoclines slope downward because each species reduces the density the other can maintain.

These isoclines produce four basic outcomes ([Vandermeer and Boucher, 1978](#)). In facultative mutualisms, both species can persist without the interaction, so both intercepts are positive. When mutualistic effects are weak relative to intraspecific limitation, the isoclines intersect at a stable equilibrium (Figure 1A), where both species coexist at higher densities

than either would reach alone. As mutualistic effects strengthen, the zero-growth isoclines become steeper, meaning that each species' zero-growth density responds more strongly to partner density. Sufficiently strong positive feedbacks eliminate the finite stable equilibrium and generate unbounded growth in the linear model (Figure 1B). This prediction contributed to historical skepticism about the biological realism of early mutualism theory (May, 1976; Boucher et al., 1982), but the fixation on unbounded growth missed the more useful interpretation: the linear model correctly identifies mutualistic feedback as a mechanism that can promote persistence at high partner abundance, even though it lacks the biological limits needed to bound that abundance.

The same graphical framework also introduces obligate mutualists, which cannot maintain positive growth in the absence of their partner. This dependence is represented by a zero or negative intercept (Figure 1C–D). When mutualistic effects are too weak to compensate for this demographic deficit, the partners cannot rescue one another and the system collapses to extinction (Figure 1C). When obligate mutualists interact strongly enough, the isoclines intersect and create a threshold point (Figure 1D). Above this threshold, each species is abundant enough to provide the benefits needed by its partner, driving both species toward higher densities. Below the threshold, one or both species are too rare to provide sufficient benefit, and the system collapses toward extinction. Although unbounded growth above the threshold is an artifact of the linear-benefit assumption, the threshold itself is not. Low-density thresholds are retained in more realistic models with saturating benefits and are one of the dynamical signatures of obligate mutualism (Hale and Valdovinos, 2021).

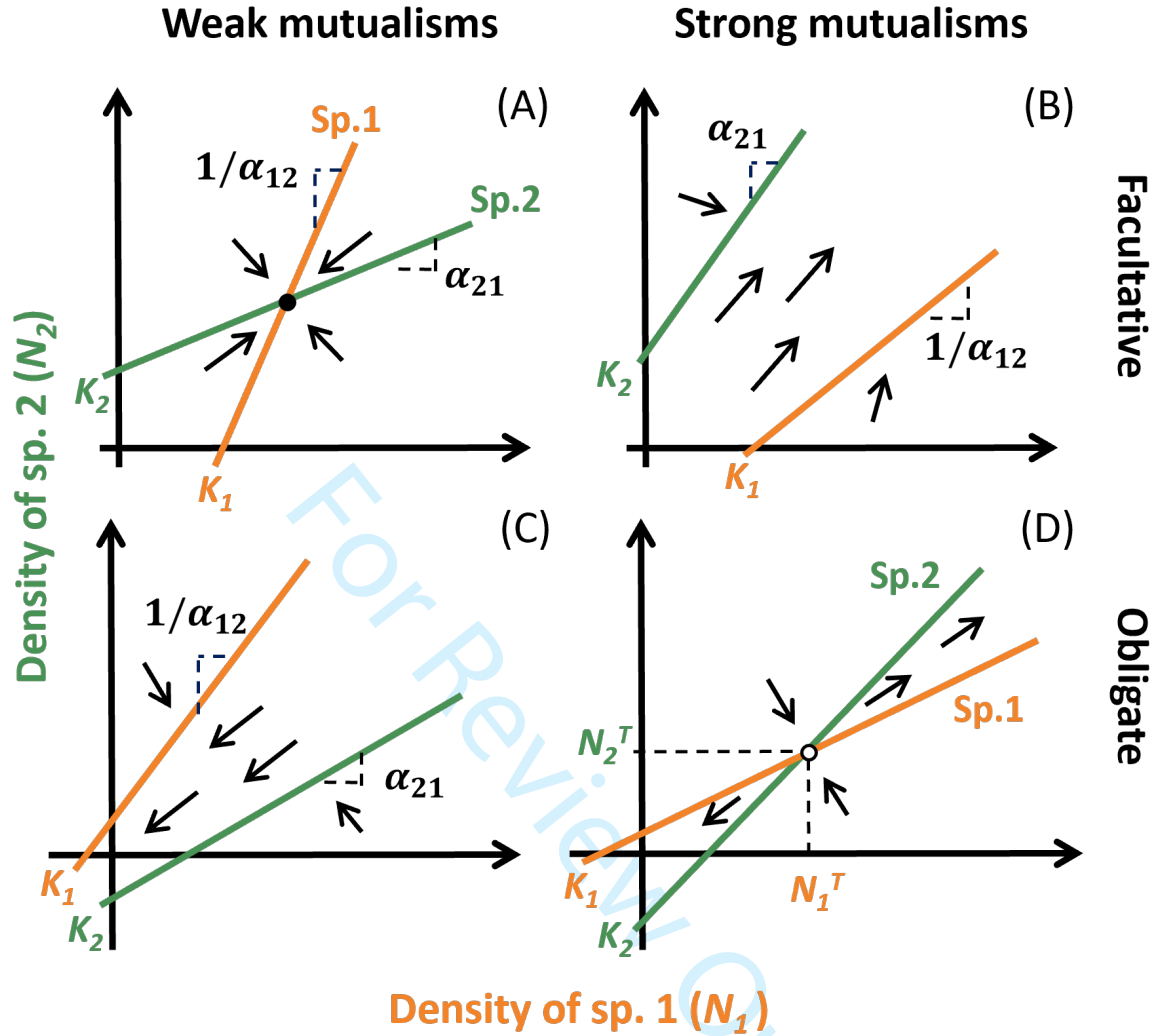


Figure 1: Outcomes of the Lotka–Volterra model of mutualism. Zero-growth isoclines provide a simple way to teach how interaction strength and partner dependence shape mutualistic dynamics. **(A)** Weak facultative mutualism: both species can persist without their partner, and weak positive effects allow the isoclines to intersect at a stable equilibrium. Both species coexist at higher densities than they would achieve alone. **(B)** Strong facultative mutualism: increasing mutualism strength, represented by larger α_{12} and α_{21} , steepens the isoclines and eliminates the stable equilibrium. In the linear Lotka–Volterra model, this produces unbounded growth. **(C)** Weak obligate mutualism: obligate mutualists have zero or negative growth in the absence of their partner, represented here by zero or negative intercepts. If mutualistic effects are too weak, partner benefits cannot compensate for this demographic deficit, and the system collapses to extinction. **(D)** Strong obligate mutualism: sufficiently strong mutualistic effects generate an unstable equilibrium that acts as a density threshold. Above this threshold, both species provide enough benefit to sustain one another; below it, the system collapses to extinction. Although the unbounded growth predicted above the threshold is an artifact of linear benefits, the threshold itself persists in more realistic models with saturating benefits. Adapted from [Vandermeer and Goldberg \(2003\)](#).

2.2 Bounding Positive Feedbacks with Saturating Benefits

The limitation of the linear Lotka–Volterra model is not that it identifies positive feedbacks, but that it allows those feedbacks to increase without bound. A logical next step is therefore to replace the linear mutualistic terms with saturating benefits. Saturation captures the biological fact that mutualistic benefits are limited by constraints such as handling time, resource production, physiological uptake, available space, or the finite demographic value of additional visits, rewards, or services. This modification changes the high-density behavior of the model without erasing its central qualitative insights: mutualists persist at high densities when partner benefits are sufficient, and obligate mutualists collapse when partner densities fall below a critical threshold (Figure 2). For example, the benefit received by species 1 from species 2, B_{12} , and by species 2 from species 1, B_{21} , can be written as:

$$B_{12} = \frac{b_{12}N_2}{h_{12} + N_2}, \quad B_{21} = \frac{b_{21}N_1}{h_{21} + N_1}, \quad (5)$$

where b_{ij} is the maximum benefit species j can provide to species i , and h_{ij} controls how quickly the benefit approaches this maximum. The model then becomes:

$$\frac{dN_1}{dt} = r_1 N_1 \left(\frac{K_1 - N_1 + \frac{b_{12}N_2}{h_{12} + N_2}}{K_1} \right) \quad (6)$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(\frac{K_2 - N_2 + \frac{b_{21}N_1}{h_{21} + N_1}}{K_2} \right). \quad (7)$$

Setting each equation to zero gives the saturating isoclines:

$$N_1 = K_1 + \frac{b_{12}N_2}{h_{12} + N_2} \quad (8)$$

$$N_2 = K_2 + \frac{b_{21}N_1}{h_{21} + N_1}. \quad (9)$$

227 These isoclines initially rise with partner density but approach the asymptotes $N_1 =$
228 $K_1 + b_{12}$ and $N_2 = K_2 + b_{21}$, respectively (dashed lines in Figure 2). Thus, saturation bends
229 the isoclines toward finite limits at high density, preventing partner-supported densities from
230 increasing without bound. In the strong facultative case, isoclines that do not intersect in
231 the linear model can intersect once benefits saturate, producing a stable finite equilibrium
232 rather than unbounded growth (Figure 2B). In the strong obligate case, saturating benefits
233 can produce two positive intersections: a lower unstable threshold and a higher stable co-
234 existence equilibrium (Figure 2D). Saturation therefore resolves the unrealistic prediction of
235 unlimited growth while preserving the low-density threshold that is characteristic of obligate
236 mutualism.

237 Interactive graphing platforms such as Desmos (Figure 2) are especially useful for
238 teaching these isoclines because students can vary α_{ij} , b_{ij} , h_{ij} , and K_i directly and immedi-
239 ately see how the isoclines shift, steepen, or bend. This makes the connection among benefit
240 functions, asymptotes, intersections, and stability more tangible than static equations alone.

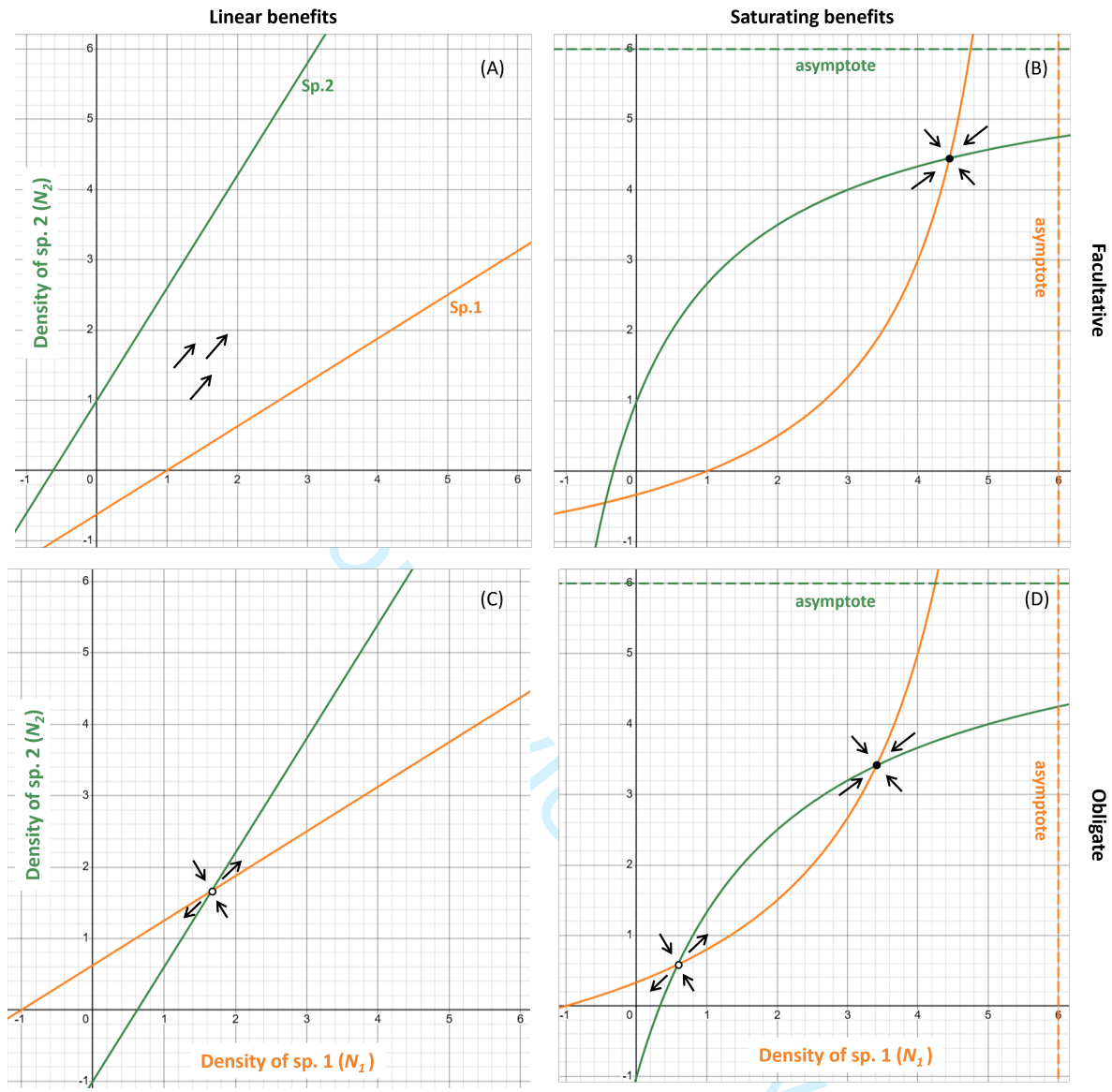


Figure 2: Saturating benefits replace unbounded growth with bounded coexistence. Orange and green curves are the zero-growth isoclines of species 1 and species 2, respectively. Dashed lines mark asymptotes. Solid circles are stable equilibria; open circles are unstable equilibria or thresholds. Arrows show the direction of density change over time. Panels A–B show strong facultative mutualism with $K_1 = K_2 = 1$; panels C–D show strong obligate mutualism with $K_1 = K_2 = -1$. Linear cases use $\alpha_{12} = \alpha_{21} = 1.6$. Saturating cases use $h_{12} = h_{21} = 2$, with $b_{12} = b_{21} = 5$ for the facultative case and $b_{12} = b_{21} = 7$ for the obligate case. **(A)** Linear strong facultative mutualism lacks a positive equilibrium and predicts unbounded growth. **(B)** Saturating benefits bend the isoclines toward finite upper bounds and create a stable coexistence equilibrium. **(C)** Linear strong obligate mutualism has an unstable threshold; above it, densities increase without bound, and below it, populations collapse to extinction. **(D)** Saturating benefits produce a lower unstable threshold and a higher stable coexistence equilibrium. Desmos graphs, in which linear and saturating nullclines can be toggled on or off, are available for strong facultative mutualism at <https://www.desmos.com/calculator/okkodn8lcd> and strong obligate mutualism at <https://www.desmos.com/calculator/qaxmnxluwl>.

2.3 Consumer–Resource Mechanisms in Mutualism Theory

Saturating-benefit models show how bounded positive effects can produce stable, finite co-existence, but they do not explain what biological processes generate the benefit function. Consumer–resource theory provides such mechanisms for mutualisms involving exchanges of resources for resources or services (Holland et al., 2005; Bronstein, 2015; Hale and Valdivinos, 2021). These include the three categories described by Bronstein (2015): nutritional mutualisms, involving resources such as carbon, nitrogen, phosphorus, or photosynthates; transportation mutualisms, involving rewards such as nectar, pollen, fruit pulp, or elaiosomes in exchange for pollen or seed movement; and protection mutualisms, involving food rewards in exchange for defense against natural enemies. Framing these interactions as consumer–resource processes also brings concepts such as functional responses, resource supply, depletion, conversion efficiency, and numerical responses into mutualism theory.

Two modeling axes, however, have been conflated in the literature (e.g., Holland and DeAngelis, 2010): consumer–resource mechanisms and benefit–cost accounting. The consumer–resource axis describes how individuals interact: one partner consumes, uses, or receives a resource or service supplied by the other. The benefit–cost axis describes the demographic consequences of that interaction: whether it increases or decreases survival, reproduction, or population growth. These axes can be combined, but they are not equivalent. A model can include consumer–resource mechanisms without explicit cost terms, or benefits and costs without tracking resource production, renewal, or depletion.

This distinction matters for interpreting Holland and DeAngelis (2010), whose model is framed as consumer–resource theory but whose distinctive nullcline geometry arises from how resource-supply costs are modeled. In its unidirectional version, species N_1 supplies a resource and receives a service, while N_2 consumes the resource and provides the service:

$$\frac{dN_1}{dt} = N_1 \left[r_1 + c_1 \left(\frac{\alpha_{12}N_2}{h_2 + N_2} \right) - q_1 \left(\frac{\beta_1N_2}{e_1 + N_1} \right) - d_1N_1 \right], \quad (10)$$

$$\frac{dN_2}{dt} = N_2 \left[r_2 + c_2 \left(\frac{\alpha_{21}N_1}{h_1 + N_1} \right) - d_2N_2 \right]. \quad (11)$$

Here, N_i is the density of mutualist i , r_i is its intrinsic growth rate, and d_iN_i represents self-limitation. The terms multiplied by c_i represent positive demographic effects: species N_1 benefits from the service provided by N_2 , and species N_2 benefits from the resource supplied by N_1 . The term multiplied by q_1 is the cost to N_1 of supplying that resource. The parameters α_{ij} and β_i scale benefits and costs, respectively, while h_i and e_i determine how those terms change with density. In the bidirectional version, the first equation is mirrored for the second species, so both partners receive saturating benefits and pay density-dependent supply costs.

This formulation motivates a consumer–resource interpretation, but the model does not track resource abundance as a state variable. Instead, resource-mediated effects are represented directly as functions of mutualist densities. Thus, Holland and DeAngelis ask how partner abundance translates into demographic benefits and supply costs, rather than modeling the resource itself. This distinction helps clarify the nullcline geometry (Figure 3).

The positive term,

$$\frac{\alpha_{12}N_2}{h_2 + N_2}, \quad (12)$$

has the same form as earlier saturating benefit functions: benefit increases with partner density and approaches an upper limit. This term curves the nullcline toward a finite asymptote, as in saturating Lotka–Volterra models. The loop-like nullclines in Holland and DeAngelis arise from the supply-cost term,

$$\frac{\beta_1N_2}{e_1 + N_1}, \quad (13)$$

which increases with consumer abundance, N_2 , and is intensified when provider abundance, N_1 , is low. Because this cost depends jointly on partner and focal abundance, it can fold the nullcline (Holland and DeAngelis, 2010; Hale and Valdovinos, 2021). Thus, the model preserves the earlier saturating-benefit result of coexistence at high densities but adds a

287 route to mutualism failure: collapse can occur when consumer abundance is high relative to
288 provider abundance, so that supply costs exceed interaction benefits. This assumption may
289 be plausible when resource provision is difficult to shut down, developmentally fixed, pas-
290 sively extracted, or delayed relative to population change, but less plausible when providers
291 can rapidly regulate investment, reduce reward production, sanction ineffective partners,
292 or exclude costly partners. This limitation highlights the difference between using partner
293 abundance as a proxy for resource-mediated effects and modeling the resource itself.

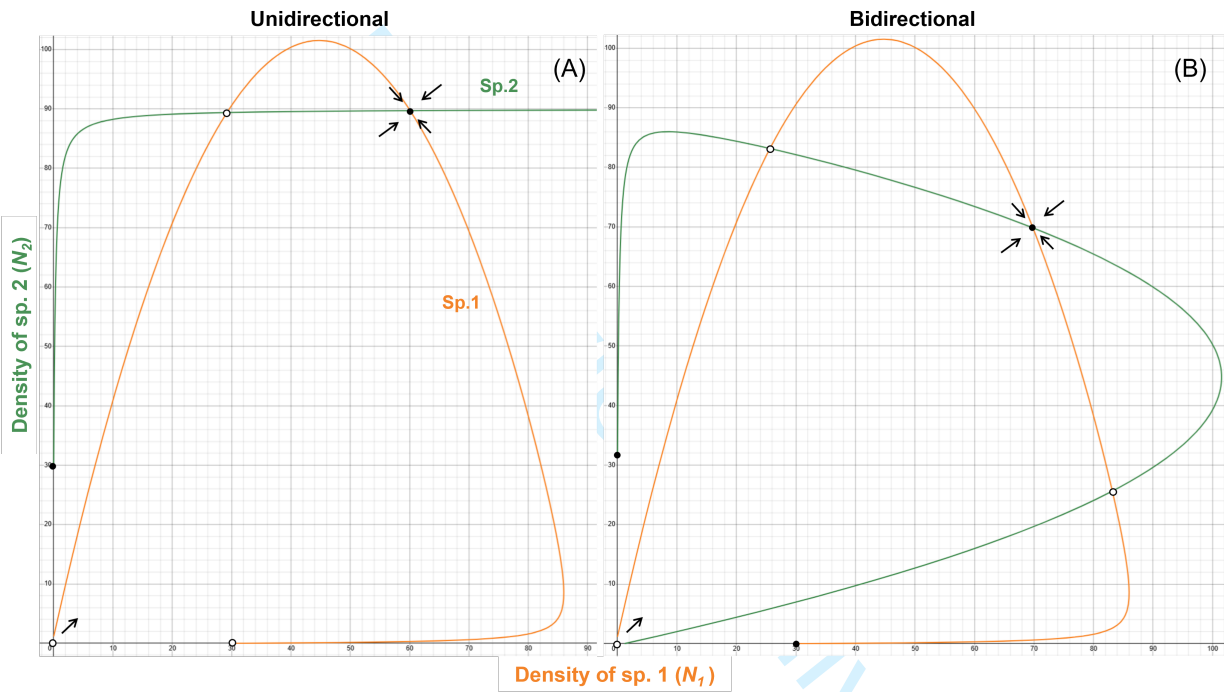


Figure 3: Cost-benefit mutualism models can produce folded nullclines and multiple equilibria. Orange and green curves are the zero-growth nullclines of facultative mutualists 1 and 2, respectively. Solid circles indicate stable equilibria, open circles indicate unstable equilibria or thresholds, and arrows show the direction of density change over time. Parameter values are $r_1 = r_2 = 0.3$, $\alpha_{12} = \alpha_{21} = 0.6$, $\beta_1 = \beta_2 = 0.2$, $c_1 = c_2 = q_1 = q_2 = 1$, $d_1 = d_2 = 0.01$, and $h_1 = h_2 = e_1 = e_2 = 0.3$. (A) In the unidirectional version, the resource-supplying species pays a density-dependent supply cost, whereas the consumer species receives a saturating benefit without an analogous cost. (B) In the bidirectional version, both partners receive saturating benefits and pay density-dependent supply costs, so both nullclines fold. Desmos graphs are available at <https://www.desmos.com/calculator/bpokvfhipg>.

294 A more explicit consumer-resource model is provided by [Valdovinos et al. \(2013\)](#), who
295 separate the resource from the abundances of the interacting species in a plant-pollinator

network. Plants produce floral rewards, pollinators consume those rewards, and pollination services feed back to plant demography. Thus, plants and animals are not linked by a single positive interaction coefficient; they are linked through resource production, visitation, reward extraction, pollen transfer, seed production, recruitment, and competition.

Floral rewards are modeled as a dynamic resource:

$$\frac{dR_i}{dt} = \beta_i p_i - \phi_i R_i - \sum_{j \in A} V_{ij} f_{ij}(R_i, p_i), \quad (14)$$

where R_i is the standing stock of rewards produced by plant species i , $\beta_i p_i$ is reward production, $\phi_i R_i$ is reward decay or self-limitation, and the final term is depletion by pollinator consumption. The visitation rate linking animal species j to plant species i is

$$V_{ij} = \alpha_{ij} \tau_{ij} a_j p_i, \quad (15)$$

where a_j is pollinator density, p_i is plant density, α_{ij} is the foraging effort of pollinator j on plant i , and τ_{ij} is visitation efficiency. Pollinator growth depends on extracted rewards, whereas plant growth depends on effective pollination, seed production, recruitment, and competition rather than pollinator abundance alone. Thus, visitation, reward consumption, reproductive service, and demographic response are separated rather than collapsed into a single positive interaction term (Valdovinos et al., 2013).

This decomposition matters because high visitation need not imply strong demographic benefit. Visits may be ineffective, rewards may be depleted, conspecific pollen may be diluted across plant species, or recruitment may be limited by post-pollination competition. By allowing pollinators to adjust foraging effort in response to reward availability, Valdovinos et al. (2013) show how animal behavior, resource depletion, and service quality scale up to species persistence and network robustness. This model therefore shifts mutualism theory from phenomenological positive effects toward explicit mechanisms linking resource production, consumption, service delivery, and population dynamics (Valdovinos

et al., 2013; Valdovinos, 2019).

Revilla (2015) extends this mechanistic logic analytically by showing how mutualistic numerical responses can be derived from resource dynamics. For a resource produced by species i and consumed by species j , Revilla writes:

$$\frac{dR_i}{dt} = \alpha_i N_i - \omega_i R_i - \beta_j R_i N_j. \quad (16)$$

If resource dynamics are fast relative to population dynamics, the resource approaches the quasi-steady state

$$R_i^* = \frac{\alpha_i N_i}{\omega_i + \beta_j N_j}. \quad (17)$$

Substituting this expression into the population equations yields numerical responses whose parameters correspond to measurable rates of resource production, loss, consumption, and conversion (Revilla, 2015). In resource-for-resource mutualisms, this approach links benefits to competition among consumers for finite rewards. In resource-for-service mutualisms, it shows how saturating service benefits can emerge from finite production and turnover of rewards such as flowers, nectar, fruits, or seeds. Together, Valdovinos-type plant–pollinator models and Revilla’s time-scale approach show how saturation, competition, and thresholds can arise from explicit resource production, depletion, service delivery, and demographic conversion.

3 Separating reciprocal benefit from facilitation and symbiosis

The framework developed above treats mutualism as a population-dynamic interaction characterized by reciprocal net benefits. This focus requires distinguishing mutualism from two related but non-equivalent concepts that are often grouped with it: facilitation and symbiosis. Mutualism occupies the $+/+$ region of the interaction spectrum, where both species

experience positive net effects (Boucher et al., 1982; Bronstein, 1994, 2015). Facilitation describes a positive effect of one species on another, often through modification of the physical or biotic environment, without specifying the reciprocal effect on the facilitator (Bertness and Callaway, 1994; Bruno et al., 2003; Callaway, 2007; Bronstein, 2015). Symbiosis describes physically intimate or persistent associations between species, regardless of whether their net outcome is mutualistic, commensal, or antagonistic (Wilkinson, 2001; Douglas, 2010; Bronstein, 2015). Thus, some facilitative interactions are mutualistic, but many are not; likewise, some mutualisms are symbiotic, but many symbioses are not mutualistic.

The distinction between mutualism and facilitation becomes especially important in environmental-gradient theory, where “positive interactions” often refers to one-way amelioration of stress rather than reciprocal demographic benefit. The stress-gradient hypothesis predicts that facilitation should become more common under high abiotic stress, whereas competition should dominate under more benign conditions (Bertness and Callaway, 1994; Callaway, 2007). Later refinements show that this pattern depends on both the nature of the stressor and the traits of the interacting species (Brooker and Callaghan, 1998; Maestre et al., 2009). When stress is caused by a shared limiting resource, such as water or nutrients, neighbours may intensify competition unless they increase access to that resource. In contrast, when stress is non-resource based, such as temperature, wind, or salinity, neighbours may facilitate persistence by ameliorating harsh conditions (Callaway, 2007; Maestre et al., 2009). Thus, environmental stress may increase positive effects without necessarily producing mutualism, because the reciprocal demographic consequences for the facilitator may remain neutral, negative, or unmeasured.

This environmental-gradient perspective is related to, but distinct from, the density-dependent arguments developed above. Stress can reduce population density, but environmental conditions can also alter interaction outcomes independently of density by changing the value of benefits, the costs of providing them, or the sign of the interaction itself (Bronstein, 1994; Maestre et al., 2009). For example, a neighbour may be beneficial under thermal

or desiccation stress but competitive when water or nutrients are limiting (Callaway, 2007; Maestre et al., 2009). Similarly, a mutualist may provide a larger demographic benefit when the exchanged resource or service is limiting, but that benefit may diminish or become costly when environmental conditions change (Hoeksema and Bruna, 2000; Heath and Tiffin, 2007; Bronstein, 2015).

For mutualism, these distinctions clarify why environmental context should be treated as a separate axis of population-dynamic variation. Mycorrhizal fungi may be most beneficial to plants under nutrient limitation, defensive mutualists may matter most under high enemy pressure, and pollination mutualisms may shift in outcome as floral rewards, pollinator abundance, or alternative resources change (Heil et al., 2001; Fellbaum et al., 2012; Cervantes-Loreto et al., 2021; Bronstein, 2015). At the same time, mutualists can compete for shared partners or limiting resources (Levin, 1972; Waser, 1978; Rathcke, 1983). Mutualism and competition are therefore not alternative explanations, but interacting processes whose balance can shift across environmental gradients. The focus of this synthesis is therefore not on all positive or intimate interactions, but specifically on the population dynamics of reciprocal net benefit.

4 Testing the Population-Dynamic Predictions of Mutualism Theory

The models discussed above yield five recurring population-dynamic predictions shared across many models of mutualism (Table 2) (Hale and Valdovinos, 2021). Mutualisms increase the densities that partners can maintain relative to their expected densities without partner benefits. Stable coexistence arises when positive feedbacks are bounded by self-limitation, finite resources, environmental constraints, partner handling limits, or service saturation. Obligate mutualisms generate low partner-density thresholds: below a critical partner density, benefits are insufficient to offset mortality, interaction costs, or demographic

dependence. Mutualisms can also generate Allee effects when focal species are too rare to attract, maintain, or receive sufficient service from their partners. At the other end of the density range, mutualisms can generate upper cost-driven thresholds when benefits saturate but the costs of supporting partners continue to rise at high or imbalanced partner densities. Together, these predictions define a bounded region of stable coexistence, with collapse possible below lower density thresholds or above upper cost-driven thresholds.

Stochasticity qualifies these deterministic predictions. Stochastic models of mutualism show that environmental or demographic fluctuations can shift systems among extinction, stochastic permanence, persistence in time average, and stationary variation around positive equilibria ([Liu and Wang, 2013](#); [Guo et al., 2023](#)). In plant-pollinator networks, stochasticity can generate extinctions even when deterministic dynamics are stable; this effect is strongest for specialists because they depend on fewer mutualist partners ([Dumoulin and Armsworth, 2022](#)). Thus, deterministic mutualism models identify demographic thresholds created by partner dependence, whereas stochasticity helps determine whether and when populations cross those thresholds.

Table 2: Empirical roadmap for population-dynamic predictions of mutualism theory

Prediction	Mechanism	Empirical test and target metric
Bounded coexistence	Benefits are limited by self-limitation, finite resources, or environmental constraints.	Add partners or resources across a gradient and test whether focal growth saturates. Estimate the density or resource level where additional input no longer increases growth.
Service or reward saturation	Partners have finite handling, search, satiation, or processing capacity.	Measure per-capita service or reward extraction across density gradients. Estimate the functional-response shape, handling time, or maximum service rate.
Low partner-density threshold	Partner rarity fails to offset mortality or interaction costs, especially in obligate mutualisms.	Reduce partner abundance or access and measure focal growth or fitness. Estimate the partner density below which growth falls below replacement.
Mutualism-induced Allee effect	Focal rarity limits partner attraction, maintenance, or receipt of sufficient service.	Track growth across gradients of focal density, visitation, reward production, or partner attraction. Estimate the focal density below which partner-mediated benefits cannot sustain growth.
Upper cost-driven threshold	Benefits saturate while costs continue to rise at high or imbalanced partner density.	Increase partner density or exploitation and measure net demographic effects. Estimate the partner density where the net effect becomes neutral or negative.

As a pedagogical roadmap, Table 2 links these theoretical predictions to measurable quantities such as partner density, resource availability, service delivery, demographic response, and per-capita population growth. Converting these quantities into rigorous tests, however, requires system-specific experimental design, appropriate null models, controls for alternative mechanisms, and attention to the temporal scale over which ecological and evolutionary responses occur. This difficulty increases for mechanistic mutualism models, which often add parameters for benefits, costs, saturation, and resource dynamics beyond those in simple Lotka–Volterra models. Empirical tests should therefore prioritize the response functions and threshold quantities that distinguish among qualitative predictions, rather than attempting to estimate every parameter in a full mechanistic model. This challenge is not unique to mutualism: debates over competition theory in the early 1980s, for example, showed that even comparatively simple and well-developed ecological concepts can be diffi-

cult to translate into decisive tests of presence, absence, or relative importance (Schoener, 1983; Roughgarden, 1983; Quinn and Dunham, 1983; Simberloff, 1983; Connell, 1983). For mutualism, the value of population-dynamic theory is therefore not to provide simple proof-or-disproof criteria, but to specify the mechanisms, response functions, threshold quantities, and null expectations needed to interpret empirical patterns.

Several existing empirical approaches already approximate the kinds of tests summarized in Table 2. Floral-array, pollen-supplementation, and visitation experiments can estimate whether service delivery or plant reproductive output saturates with partner or reward availability (Harder, 1990; Mitchell and Waser, 1992; Morris et al., 2010). Partner-removal experiments in protective mutualisms, such as classic ant–Acacia manipulations, show how reduced defender abundance can affect herbivory, growth, and mortality, and could be extended from binary removals (Janzen, 1966) to density gradients that estimate lower thresholds. Finally, isotopic or resource-tracing approaches in mycorrhizal mutualisms can quantify the currencies exchanged between partners, allowing researchers to test whether costs continue to rise after nutritional or demographic benefits saturate (Kiers et al., 2011; Fellbaum et al., 2012). The key shift is not inventing entirely new methods, but using existing manipulations to estimate demographic response functions and threshold densities.

These examples also illustrate why empirical tests of mutualism theory must be interpreted at the appropriate temporal scale. Low partner-density thresholds describe contemporary demographic conditions under which an established interaction can persist; they are not, by themselves, a theory for the historical origin of obligacy (Bronstein, 2015). Obligate mutualisms may avoid initial low-density collapse through spatial aggregation, sequential establishment, environmental subsidies, facultative early stages, or local conditions that allow one partner to persist before full reciprocal dependence evolves. Similarly, long experiments may measure eco-evolutionary outcomes rather than purely ecological responses, especially when partners differ strongly in generation time, as in legume–rhizobium interactions (Heath and Tiffin, 2007; Simonsen and Stinchcombe, 2014). Thus, Table 2 identifies measurable de-

mographic targets, but interpreting those targets requires attention to establishment history, ecological context, and the timescale over which partner traits can change.

5 Discussion

This forum piece began with two explanations, often invoked implicitly or explicitly, for why mutualism has remained peripheral to ecological theory: the unbounded growth predicted by early Lotka–Volterra mutualism models (May, 1976, 1982), and the assumption that the diversity of mechanisms by which species benefit one another defies theoretical unification (Risch and Boucher, 1976; Boucher et al., 1982; Bronstein, 1994). Neither explanation is sufficient to keep mutualism outside the core teaching of population ecology. The problem with the Lotka–Volterra model is not that mutualism is inherently unstable, but that linear benefits increase without bound. When benefits saturate, mutualisms can produce finite coexistence while retaining the low-density thresholds that make obligate positive interactions dynamically distinctive. Likewise, the diversity of mutualistic mechanisms does not preclude synthesis. Pollination, seed dispersal, protection, nutritional exchange, and mycorrhizal mutualisms differ biologically, but they can be connected through common population-dynamic features: partner dependence, benefit saturation, resource production, service delivery, interaction costs, and density-dependent thresholds.

The population-dynamic synthesis developed here from existing mutualism theory matters because it changes what empiricists might measure. If mutualism is taught mainly through examples, then empirical work will naturally emphasize species identities, natural history, or interaction occurrence. The population-dynamic synthesis developed here from existing mutualism theory matters because it changes what empiricists might measure. If mutualism is taught mainly through examples, then empirical work will naturally emphasize species identities, natural history, or interaction occurrence. Those data are valuable, but they mostly describe who interacts with whom and how (Bronstein, 1994) rather than how

those interactions change per-capita growth, persistence, or stability. A population-dynamic framing instead directs attention to per-capita growth rates, partner-density gradients, functional responses, benefit–cost curves, and the densities at which interactions shift from positive to neutral or negative. Table 2 is meant to support this transition. It does not prescribe a single experimental design, but it shows how theoretical predictions can be translated into measurable demographic targets.

Another common reason for treating competition as more foundational than mutualism is that competition follows directly from shared limiting factors: when species depend on the same finite resource, enemy-free space, host, partner, or service, asymmetries in access or use can generate competitive effects (Tilman, 1982; Chesson, 2000; Chase and Leibold, 2003). This logic also applies within mutualisms. Pollinators can compete for floral rewards, plants can compete for pollinator service, and mycorrhizal fungi can compete for host carbon or soil nutrients (Levin, 1972; Waser, 1978; Rathcke, 1983; Bever et al., 2010; Kiers et al., 2011; Fellbaum et al., 2012). Yet the same observation points beyond a competition-centered view: mutualistic partners can themselves be limiting factors. Partner abundance, partner quality, resource exchange, or service delivery can determine whether a focal population maintains positive growth (Bronstein, 1994; Holland and DeAngelis, 2010; Peay, 2016; Hale and Valdovinos, 2021). Thus, competition can occur within mutualisms, but mutualism can also alter the conditions under which competition, resource limitation, succession, niche construction, and coexistence occur (Bruno et al., 2003; Bimler et al., 2018; Peay, 2016; Koffel et al., 2021).

The mutualistic niche framework makes this broader implication explicit: mutualism can help define the niche conditions under which populations persist (Peay, 2016; Koffel et al., 2021). Traditional niche theory often treats abiotic tolerances and resource requirements as intrinsic properties of species, with biotic interactions modifying where those species are realized. Peay (2016)’s distinction between the conditions under which a species can persist without biotic interactions and those under which it can persist with mutualist partners

shows why niche theory need not be competition-centered. Mycorrhizal fungi, for example, can lower plant resource requirements, alter access to nutrient pools, and shift both requirement and impact niches. Mutualism therefore does not merely operate after the niche is defined; it can define the demographic conditions that make persistence possible.

The marginal position of mutualism and other positive interactions in ecological training may also discourage the development and uptake of new theory. When a field has widely taught models, standard empirical systems, and shared analytical tools, new theoretical extensions are more likely to be used, tested, cited, and recognized as broadly relevant. Competition theory benefits from this infrastructure: new tools can be connected to a large community already trained to interpret them. Positive interactions lack comparable infrastructure. As a result, developing theory for facilitative or mutualistic communities can carry lower expected professional returns, not because the questions are unimportant, but because fewer empirical datasets, model systems, and pedagogical conventions are available to make the theory immediately useful. In this sense, the limited integration of positive interactions can become self-reinforcing: because fewer researchers are trained to use the theory, new theoretical tools may attract less attention and be slower to influence empirical practice. Building a common population-dynamic framework for mutualism is therefore not only a teaching goal; it is also a way to make future theoretical and empirical work on positive interactions more cumulative.

The task, then, is not to invent mutualism theory from scratch, but to make existing theory more central to ecological pedagogy and empirical practice. The argument is not that mutualism should replace competition as an organizing principle, but that it shapes many of the same ecological objects that competition theory has long emphasized: limiting factors, niches, zero-growth boundaries, equilibria, stability, and coexistence. Teaching mutualism through equations, graphical analysis, demographic response functions, and empirical tests need not reduce the natural-history richness of mutualistic systems. Instead, it can connect that richness to general principles that can be modeled, visualized, and tested.

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